EXP 9:Write the python to implement Travelling Salesman Problem

```
from itertools import permutations

graph = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

n = len(graph)
min_cost = float('inf')

for path in permutations(range(1, n)):
    route = (0,) + path + (0,)
    cost = 0

    for i in range(len(route) - 1):
        cost += graph[route[i]][route[i + 1]]

    if cost < min_cost:
        min_cost = cost
        best_path = route

print("Best path:", best_path)
print("Minimum cost:", min_cost)
```

Output

```
A B D E C F Best path: (0, 1, 3, 2, 0)
Minimum cost: 80
```